

Assessment 2 — Multiple Choice

Information Technology 10 (IT 10) • Finding Resistance Value & Resistance Color Code • 20 items • Answer key included

Learning Area	Information Technology 10 (TLE / ICT — Electronics)	Test Type	Multiple Choice
Content / Topic	Reading 4-band resistor color codes: finding resistance value and finding the color code	No. of Items	20
Date	_____	Rating	____ / 20

Table of Specification (TOS)

Content / Topic	Learning Competency	No. of Items	%	Cognitive Level		
				Remember	Understand	Apply
A. Finding the Resistance Value from the Color Bands (4-band resistors, given bands → value)	Interprets the digit and multiplier bands of a 4-band resistor to compute its resistance in ohms (Ω), kilohms (k Ω), or megohms (M Ω).	10	50%	2	3	5
B. Finding the Resistance Color Code from the Value (given value → bands)	Converts a given resistance value into its correct 4-band color code (two significant digits, multiplier, tolerance).	10	50%	2	3	5
TOTAL		20	100%	4	6	10
Percentage per cognitive level				20%	30%	50%

Reference: Standard 4-band resistor color code (IEEE / IEC 60062).

Directions: Read each item carefully and write the letter of the correct answer on the space provided. (1 point each.)

Quick Reference — 4-Band Resistor Color Code

Color	Black	Brown	Red	Orange	Yellow	Green	Blue	Violet	Grey	White	Gold	Silver
Digit	0	1	2	3	4	5	6	7	8	9	—	—
Multiplier	×1	×10	×100	×1k	×10k	×100k	×1M	×10M	×100M	×1G	×0.1	×0.01
Tolerance	—	±1%	±2%	—	—	±0.5%	±0.25%	±0.1%	±0.05%	—	±5%	±10%

Part I — Finding the Resistance Value (Items 1-10)

From the given color bands, determine the resistance value of the 4-band resistor.

- Brown – Black – Red – Gold
A. 100 Ω B. 1 k Ω C. 10 k Ω D. 100 k Ω
- Red – Red – Brown – Gold
A. 22 Ω B. 220 Ω C. 2.2 k Ω D. 22 k Ω

3. Yellow – Violet – Orange – Gold
A. 470 Ω B. 4.7 k Ω C. 47 k Ω D. 470 k Ω
4. Green – Blue – Brown – Gold
A. 56 Ω B. 560 Ω C. 5.6 k Ω D. 56 k Ω
5. Brown – Black – Orange – Gold
A. 1 k Ω B. 10 k Ω C. 100 k Ω D. 1 M Ω
6. Orange – Orange – Red – Gold
A. 330 Ω B. 3.3 k Ω C. 33 k Ω D. 330 k Ω
7. Brown – Black – Green – Gold
A. 100 k Ω B. 1 M Ω C. 10 M Ω D. 100 M Ω
8. White – Brown – Yellow – Gold
A. 91 k Ω B. 910 k Ω C. 9.1 M Ω D. 91 M Ω
9. Blue – Grey – Red – Gold
A. 68 Ω B. 680 Ω C. 6.8 k Ω D. 68 k Ω
10. Orange – White – Brown – Gold
A. 39 Ω B. 390 Ω C. 3.9 k Ω D. 39 k Ω

Part II — Finding the Resistance Color Code (Items 11-20)

From the given resistance value, determine the correct 4-band color code (all items use gold $\pm 5\%$ tolerance).

1. 220 Ω
A. Red — Red — Brown — Gold B. Red — Red — Red — Gold
C. Brown — Red — Red — Gold D. Orange — Orange — Brown — Gold
2. 1 k Ω (1,000 Ω)
A. Brown — Black — Brown — Gold B. Brown — Black — Red — Gold
C. Red — Black — Brown — Gold D. Brown — Black — Orange — Gold
3. 4.7 k Ω (4,700 Ω)
A. Yellow — Violet — Brown — Gold B. Yellow — Violet — Red — Gold
C. Violet — Yellow — Red — Gold D. Yellow — Violet — Orange — Gold
4. 10 k Ω (10,000 Ω)
A. Brown — Black — Red — Gold B. Brown — Black — Orange — Gold
C. Brown — Black — Yellow — Gold D. Orange — Black — Brown — Gold
5. 330 Ω
A. Orange — Orange — Brown — Gold B. Orange — Orange — Red — Gold
C. Red — Orange — Brown — Gold D. Orange — Orange — Orange — Gold
6. 470 Ω
A. Yellow — Violet — Brown — Gold B. Yellow — Violet — Red — Gold
C. Violet — Yellow — Brown — Gold D. Yellow — Violet — Orange — Gold

7. 100 Ω

- A. Brown — Black — Black — Gold B. Brown — Black — Brown — Gold
C. Brown — Black — Red — Gold D. Red — Black — Brown — Gold

8. 6.8 k Ω (6,800 Ω)

- A. Blue — Grey — Brown — Gold B. Blue — Grey — Red — Gold
C. Grey — Blue — Red — Gold D. Blue — Grey — Orange — Gold

9. 3.3 k Ω (3,300 Ω)

- A. Orange — Orange — Brown — Gold B. Orange — Orange — Orange — Gold
C. Orange — Orange — Red — Gold D. Red — Orange — Orange — Gold

10. 47 k Ω (47,000 Ω)

- A. Yellow — Violet — Red — Gold B. Yellow — Violet — Orange — Gold
C. Violet — Yellow — Orange — Gold D. Yellow — Violet — Yellow — Gold

Answer Key — Assessment 2

No.	Answer	Solution	No.	Answer	Solution
1	B	Brown(1) Black(0) $\times 100 = 1,000 \Omega = 1 \text{ k}\Omega$	11	A	220 = 2-2- $\times 10 \rightarrow$ Red, Red, Brown, Gold
2	B	Red(2) Red(2) $\times 10 = 220 \Omega$	12	B	1,000 = 1-0- $\times 100 \rightarrow$ Brown, Black, Red, Gold
3	C	Yellow(4) Violet(7) $\times 1\text{k} = 47,000 \Omega = 47 \text{ k}\Omega$	13	B	4,700 = 4-7- $\times 100 \rightarrow$ Yellow, Violet, Red, Gold
4	B	Green(5) Blue(6) $\times 10 = 560 \Omega$	14	B	10,000 = 1-0- $\times 1\text{k} \rightarrow$ Brown, Black, Orange, Gold
5	B	Brown(1) Black(0) $\times 1\text{k} = 10,000 \Omega = 10 \text{ k}\Omega$	15	A	330 = 3-3- $\times 10 \rightarrow$ Orange, Orange, Brown, Gold
6	B	Orange(3) Orange(3) $\times 100 = 3,300 \Omega = 3.3 \text{ k}\Omega$	16	A	470 = 4-7- $\times 10 \rightarrow$ Yellow, Violet, Brown, Gold
7	B	Brown(1) Black(0) $\times 100\text{k} = 1,000,000 \Omega = 1 \text{ M}\Omega$	17	B	100 = 1-0- $\times 10 \rightarrow$ Brown, Black, Brown, Gold
8	B	White(9) Brown(1) $\times 10\text{k} = 910,000 \Omega = 910 \text{ k}\Omega$	18	B	6,800 = 6-8- $\times 100 \rightarrow$ Blue, Grey, Red, Gold
9	C	Blue(6) Grey(8) $\times 100 = 6,800 \Omega = 6.8 \text{ k}\Omega$	19	C	3,300 = 3-3- $\times 100 \rightarrow$ Orange, Orange, Red, Gold
10	B	Orange(3) White(9) $\times 10 = 390 \Omega$	20	B	47,000 = 4-7- $\times 1\text{k} \rightarrow$ Yellow, Violet, Orange, Gold

Scoring guide: 1 point per correct answer; 20 points total. In both parts, the gold band indicates $\pm 5\%$ tolerance. Formula used: $R = (1\text{st digit} \times 10 + 2\text{nd digit}) \times \text{multiplier}$.

Prepared by

IT 10 Teacher

Checked / Approved by

Department Head / Coordinator